

Student Answer Sheet

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Q1. Which blood vessel carries blood away from the heart?

Arteries carry blood away from the heart.

Q2. Which of the following organelles is primarily involved in photosynthesis? (a) Mitochondria (b) Ribosomes (c) Chloroplast

- it contains chlorophyll which captures light energy for photosynthesis.

Q3. Explain the role of chloroplasts in photosynthesis, naming the main pigments involved and briefly outline the two stages.

Chloroplasts contain chlorophyll pigments that absorb light. Photosynthesis has two stages: light-dependent reactions (in thylakoids) and Calvin cycle (in stroma).

Q4. Describe the flow of blood through the human heart starting from the right atrium and ending at the aorta.

Blood enters right atrium -> tricuspid valve -> right ventricle -> pulmonary valve -> pulmonary artery -> lungs -> pulmonary veins -> left atrium -> mitral valve -> left ventricle -> aortic valve -> aorta.

Q5. Draw a labelled diagram of an alveolus showing capillaries and air space (label alveolar sac, capillaries, and arrows showing O₂ in, CO₂ out).

[Diagram: Circle representing alveolus with squiggly lines for capillaries, arrows showing O₂ in, CO₂ out]

Student Answer Sheet (continued)

Q6. Draw a neat labelled diagram of the human digestive system (stomach, small intestine, large intestine)

[Diagram: Human digestive system with mouth, esophagus, stomach (J-shaped), small intestine (coiled), large intestine, liver (right side), pancreas (below stomach). Absorption site labelled at small intestine.]

Q7. Draw and label a nephron (Bowman's capsule, glomerulus, proximal tubule, loop of Henle, distal tubule)

[Diagram: Nephron showing Bowman's capsule (cup-shaped), glomerulus (capillary network inside), proximal tubule, loop of Henle (U-shaped), distal tubule, collecting duct (final tube).]

Q8. Explain the structural differences between palisade mesophyll and spongy mesophyll and state how they are adapted for their function.

Palisade mesophyll: cells are tall, tightly packed, and rich in chloroplasts - optimized for light absorption. Spongy mesophyll: cells are loosely arranged with air spaces - optimized for gas exchange (CO₂ and O₂).

Q9. Describe the process of transpiration in plants in two to three sentences and name two environmental factors that affect the rate.

Transpiration is the loss of water vapor from leaves through stomata. It creates a pulling force that draws water up from roots. Factors: high temperature and low humidity increase transpiration rate.

Q10. Explain how the structure of xylem vessels facilitates water transport in plants (mention one structural feature).

Xylem vessels have lignified walls that provide structural support and prevent collapse under tension. This allows continuous water column to be pulled upward from roots to leaves.

Student Answer Sheet (continued)

Q11a. A diagram shows two potted plants - Plant A in bright light with broad green leaves, Plant B kept in

Plant A has broad green leaves while Plant B has pale, elongated and stretched leaves.

Q11b. Suggest one practical measure to help Plant B recover.

Move Plant B to a brighter location with adequate sunlight so it can perform photosynthesis properly and recover its green color.

Q12. A resting person has tidal volume (air per breath) of 0.5 L and breathes 12 times per minute. Calculate

Minute ventilation = Tidal volume $\times$ Respiratory rate = $0.5 \text{ L} \times 12 \text{ breaths/min} = 6 \text{ L/min}$

Q13. If dead space is 0.15 L per breath, calculate the alveolar ventilation per minute. Show working.

Alveolar ventilation = (Tidal volume - Dead space) $\times$ Rate = $(0.5 - 0.15) \times 12 = 0.35 \times 12 = 4.2 \text{ L/min}$